



Design Proposal — White Paper

Quick Start OCPS: Smart Power AI Design Contest · Design Spring

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Part 1: City Power Profile

Select 10 building types from the all_users_energy dataset to represent your smart city. For each building type, provide its energy profile and assign it to a priority tier. Every assignment must be justified — judges will ask you to defend your choices.

1A. User Profiles

Complete the table below for your 10 selected building types.

Building Type	Mean kWh	Peak kWh	Priority Tier	Justification
1. Hospital	492.12	693.28	Critical	Life Saving and absolute backbone of a city can provide everything in an emergency.
2. Fire Station	11.53	13.7	Critical	Lifesaving first responders will be the first to respond in an emergency.
3. Police Station	10.75	13.29	Critical	Lifesaving first responders and keep order during emergency.
4. Clinic	1.79	2.56	Critical	Lifesaving and can provide hospital overflow in an emergency.
5. Grocery	72.33	114.7	Critical	Main provider of food in a community and perishables will go bad without power.
6. Gas Station	21.41	28.53	Flexible	Provides fuel for transport, important for first responders and people.
7. K12 School	63.46	108.7	Flexible	Acts as an emergency shelter in weather emergencies
8. Library	11.47	15.5	Flexible	Acts as an emergency shelter in weather emergencies
9. Fast Food	9.5	15.41	Flexible	Provides cheap food
10. Restaurant	13.71	21.55	Flexible	Provides food

1B. Labeling (Classification) Decisions

Which building type(s) was most difficult to assign to a priority tier? Explain the tension and how your team resolved it.

Describe your hardest labeling decision here...

Grocery stores were the hardest building type to classify. In an emergency they are not as immediately crucial as the other critical buildings, such as a fire station or hospital, but they



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are still crucial to survival in a community being the main source of food. We settled this by placing it in the critical tier; it is still too important to wait in the event of an emergency and should be prioritized.



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Part 2: System Data Architecture

Your AI system needs data — not just the dataset you already have, but a continuous flow of information from the real world. Map at least three data streams that would feed your smart city AI system.

2A. Data Streams

Identify at least three data streams. For each one, describe what it contains, where it comes from, and what format it is in. Add additional data streams you think would strengthen the efficacy of your AI system.

Data Stream	What it contains	Where it comes from	Format & frequency
Historical (required)	all_users_energy.csv — building energy profiles	Contest dataset	CSV, 15-min intervals, one day profiles
Real-time (required)	Weather data, power levels, current power needs.	Real world sensors, thermometers, power sensors, etc.	Live data constantly streaming to the model as an input.
Contextual (required)	Emergency event types	Human input	Data inputted through a human to computer interfaces
(optional)			

2B. Workflow Diagram

Build your AI pipeline workflow diagram in draw.io: <https://app.diagrams.net/>.

You can find tutorials on YouTube: <https://www.youtube.com/watch?v=bN6i6dsoZTs>

Your diagram should show:

data sources on the left --> models in the middle --> outputs --> actions on the right

What your diagram must show

At least 3 data sources flowing into your system

The model(s) your system uses (regression, classification, optimization)

The output of each model — what does it predict or decide?

The action taken based on each output — what does the smart city DO with that prediction?

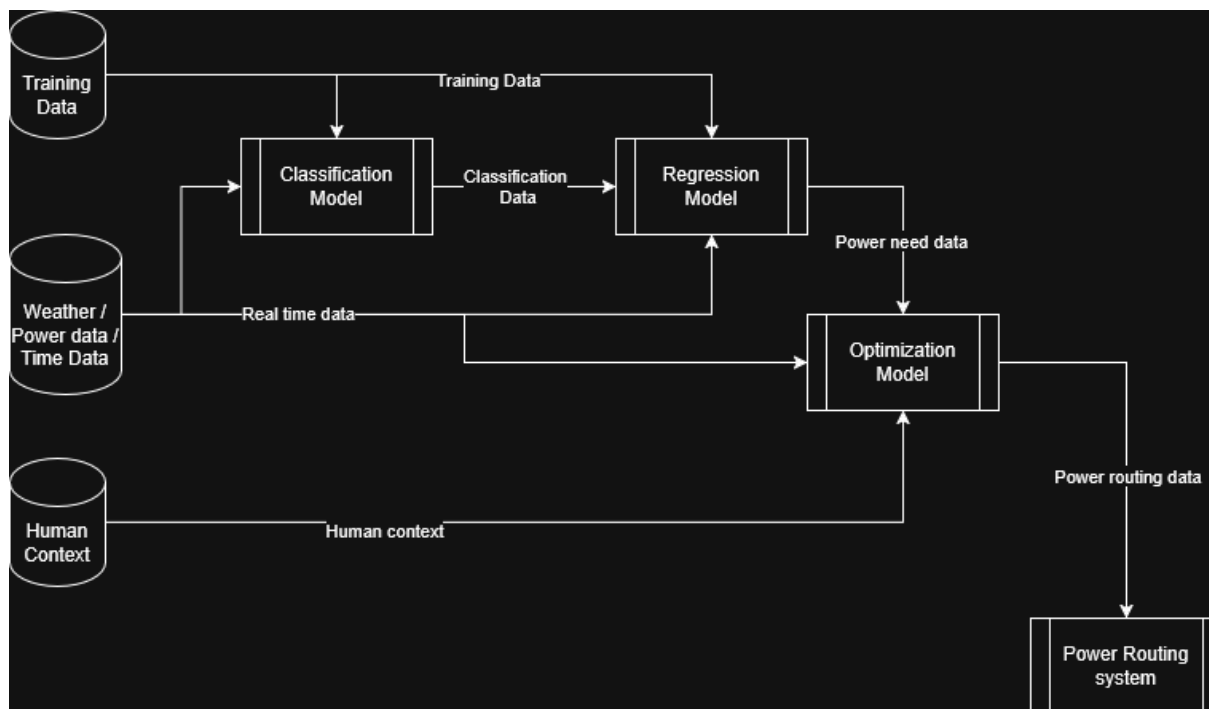
Export your draw.io diagram as a PNG and paste it below:



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Part 3: Our AI Pipeline

Describe the AI models your smart city system would use. For each model, explain what question it answers, what data it needs, and what information the model outputs. Week 1 teams can propose this conceptually. Week 2 teams should be able to reference actual results from the programming notebooks.

3A. Regression Model – Predicting Demand

Would your system use regression? If yes, what does it predict? What features does it use? What would the output look like? (Example: predicted kWh demand for the next hour for each building type.)

Week 2 teams: Include your R^2 score and MAE from the regression notebook. What did your model learn about which features matter most?

Our system would use regression. The model uses regression to predict how much power a given building will need. The model takes in current weather conditions, time of day, etc. and outputs the predicted hourly kWh demand. During training our model learned that time mattered the most when predicted power needs. Our model achieved a r^2 of 0.958 and a MAE of 25.8 kWh

3B. Classification Model – Assigning Priority

Would your system use classification? If yes, what does it classify? What tiers does it assign? What happens when a building is misclassified?

Week 3 teams: Include your confusion matrix results. Which buildings were hardest to classify? What are the consequences of those errors?

Our system would use a classification model to classify different buildings into their respective importance tiers based on their power draw and variance. It assigns buildings into three tiers, critical, flexible, and non-critical. Our model was very successful in prediction tests, only failing with gas stations, misclassifying them into non-critical instead of flexible 18% of the time. These errors would result in a percent of gas stations not being given the power they need in the event of an emergency, but only a small portion of them would go down.



3C. Optimization – Distributing Power

During an emergency, how would your system decide who gets power first? Describe the rules or algorithm your system would follow. What constraints must it respect? What parameters is the system optimizing for? All teams write this conceptually — no code required.

Our system would route power based on the previous routing decided by the first two models and human/sensor input data formed. This model would attempt to find any errors made in the previous two models and apply extra, scenario-based changes to the routing data.

Part 4: Our Evidence

What does the data tell you? **Every claim in your design proposal should be supported by evidence — a visualization, a model output, or a specific number from your analysis.**

Include at least two visualizations with written interpretation.

What good evidence looks like

NOT: 'Here is a graph of hospital energy use.'

YES: 'This graph shows that hospital energy demand varies by less than 18% throughout the day, which is why we classify it as Critical — it runs consistently at high demand around the clock.'

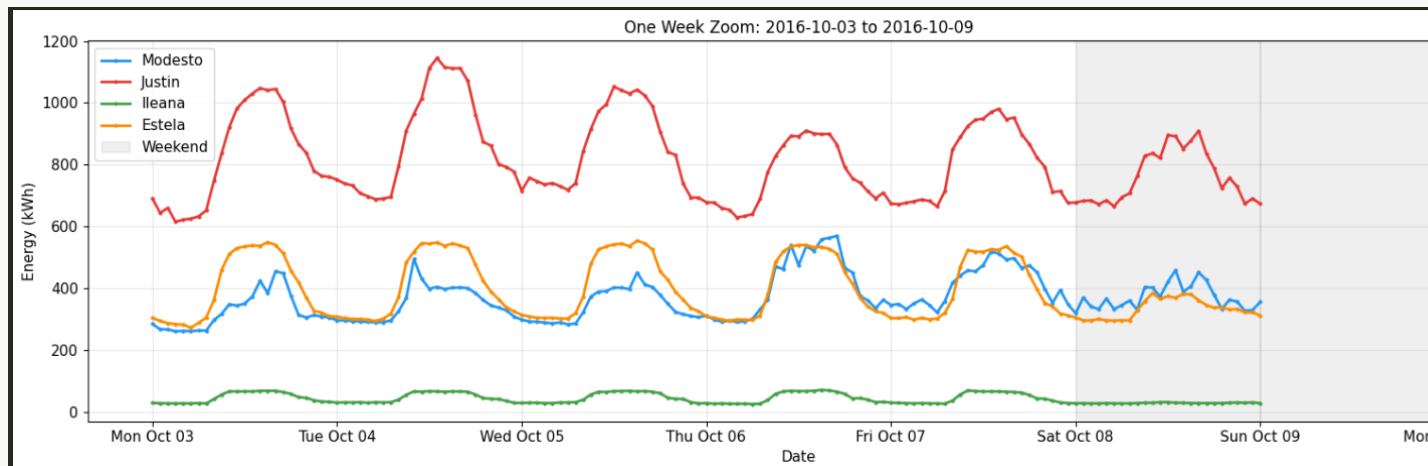
Every visualization must answer a specific question. State the question, show the evidence, interpret what it means for your design.

Visualization 1

Question this visualization answers:

How does power usage vary over time on the daily scale?

Visualization:





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What this tells us about our design:

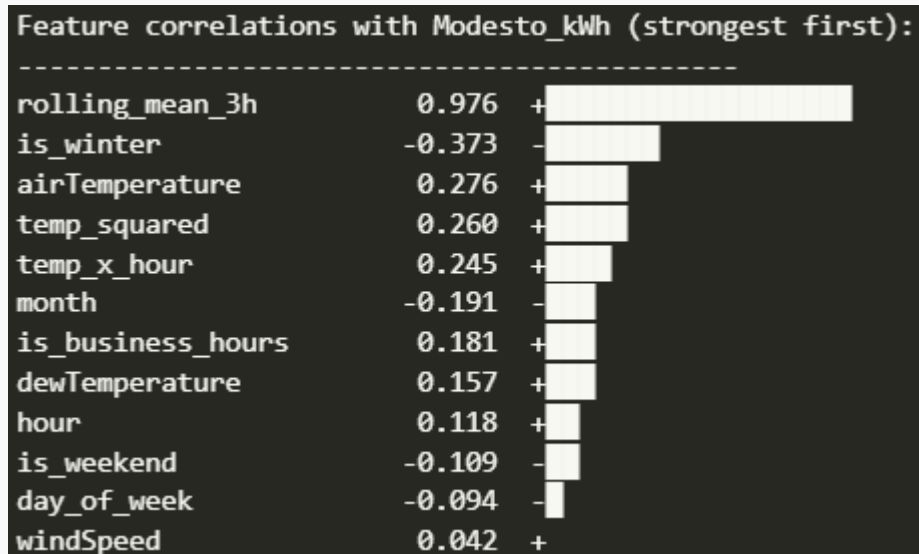
This diagram plots the power usage of each building at the hospital; it shows a large variance in power usage throughout the day and differences within different days of the week, meaning that it will be crucial to factor in day and time into our model.

Visualization 2

Question this visualization answers:

What features correlate with the buildings power usage?

Visualization:



What this tells us about our design:

This correlation chart shows different features of correlation with the Modesto building's energy usage. Excluding the rolling mean due to it being engineered off the data, the most important features appear to be the time of year and the temperature.



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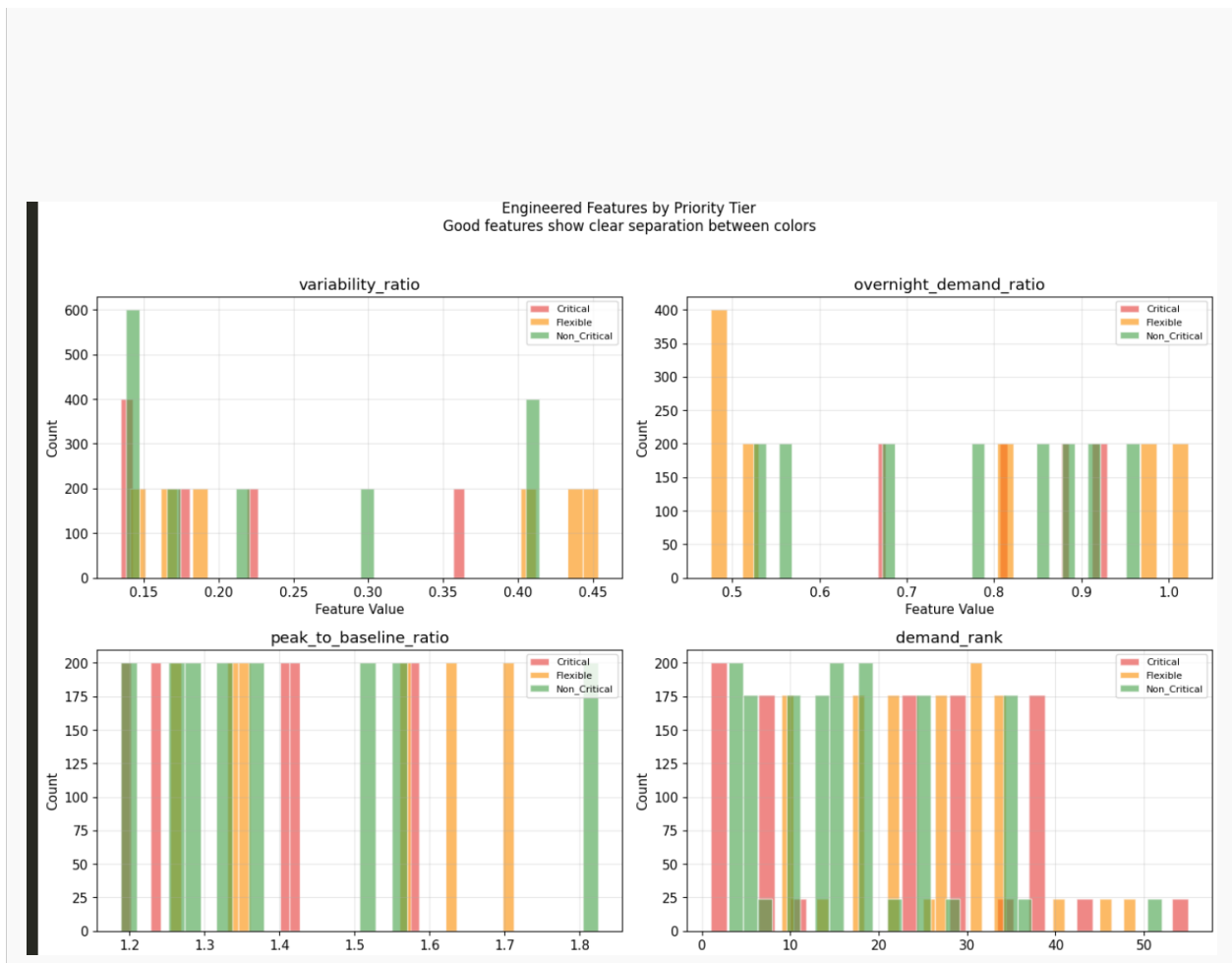
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Visualization 3 (optional — encouraged for Week 3 teams)

Question this visualization answers:

How do individual features correlate with a building's criticality ranking?

Visualization:



What this tells us about our design:

At first sight there seems to be no correlation between individual features and a buildings' criticality ranking; this demonstrates why we used a decision tree model to classify buildings. Instead of trying to find a trend in the data, it divides data in a sort of binary search to determine its ranking.



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Part 5: Design Limitations

This is the section judges will focus on. A design that knows its own limitations is more trustworthy than one that claims to do everything. Answer every prompt specifically — use numbers, building names, and scenario details from your analysis.

What this design does well

Our design performs well when:

Our designs perform well in extreme temperature scenarios, during business hours, and rerouting power in an emergency.

The data condition where our model is most accurate is:

During business hours under normal, moderate weather conditions. Since the model is trained from data that primarily consists of these conditions, this is what it can predict the best.

The community members who benefit most from our design are:

Hospital patients, medical staff, first responders, and any other citizen in an emergency. The model is designed to prioritize the most crucial power needs in the event of an emergency; in turn it benefits every party involved in these emergencies.

Where this design would struggle

Our design would struggle when:

When predicting energy for flexible buildings. In our model and selections, flexible buildings tend to be wildly spread out in all sorts of factors, primary energy consumption, and variability. These factors make it difficult for a model to create parameters that consistently identify them.

The building type our system might misclassify is gas stations because:



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Gas stations are hard to predict; they come in many different sizes and have different power requirements. Their power usage data also tends to be very close to surrounding non-critical infrastructure such as hotels.

During a hurricane scenario, the biggest risk in our design is:

Demand surges, in the event of hurricane hospitals and other emergency infrastructure, experience a surge in power use. This surge is relatively unpredictable as the majority of training data does not include such surges.

Who might be harmed

The community members most at risk from an error in our system are:

Standard citizens attempt to evacuate and get gas for backup generators along with first responders in need of gas for their emergency vehicles.

A specific scenario where our classification error could cause harm:

A hurricane comes in and knocks out most of the power, then the classification system misclassifies half of the city's gas stations. Now emergency responders and regular citizens will have to travel farther to the nearest working gas station, or if there is not a powered station within a reasonable range, they may simply have to deal with what they currently have.

How we would reduce that risk in a future version:

We would be able to reduce this risk using a third optimization model. This model would take input that the classification model can't see (human input / specific event data) and review and modify the power routing based on these inputs.

How we would improve the design

If we had three more months we would add:



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If we had three more months, we would first implement the third optimization model, taking real world and human made inputs and optimizing the power routing based on such, then we would implement more advanced machine learning techniques rather than linear regression for the regression model, potentially replacing it with a random forest model or neural network. We would then implement real world non-functional testing, meaning that the model would be running in a simulation taking real live inputs and outputting real data that is not used.

The data our system needs that we do not currently have:

Weather event data, individual building data instead of type, power usage within buildings, population data, and location data for buildings and people.

The model we would most want to improve and why:

The regression model, it currently functions using linear regression. This is good when data has very strong correlation but is highly susceptible to outliers. A random forest model would likely be best; these models use multiple decision trees for different variables (temperature, time, etc.) and average them to create a complex and nonlinear prediction model, perfect for the current data.

Prepare for Submission

Instructions: Save this document as a PDF titled **Design Proposal_[team name].pdf**. You will submit this document as part of Task 09